

Date: _____

vehicles, accepted, rejected, B, V, C, RA, RB, RC

Q5. Declare capacityA, capacityB, capacityC : INT
 Declare type: Char
 Declare category: char
 Declare permit, emergency : Boolean
 CapacityA, capacityB, capacityC, vehicles, accepted, rejected, B, V, C, RA, RB, RC ← 0
 Output "Enter type (B, C, V), category (F, S, G), Permit (Y/N), emergency (Y/N): "
 Input type, category, Permit, emergency
 while (type ≠ B OR type ≠ C OR type ≠ V) AND (category ≠ F OR category ≠ S OR category ≠ G) AND (Permit ≠ Y OR Permit ≠ N) AND (emergency ≠ Y OR emergency ≠ N)
 Do
 Output "Invalid data, enter again:"
 Input type, category, Permit, emergency
 Endwhile
 ← vehicles ← vehicles + 1
 If Permit = TRUE OR emergency = TRUE then
 If category == 'F' then
 If type == 'B' OR type == 'C' then
 If capacityA < 20 then
 capacityA ← capacityA + 1
 Output "park in zone A"
 accepted ← accepted + 1
 Else
 Output "Space full", rejected ← rejected + 1
 EndIf
 Else if type == 'V' then
 If capacityA < 19 then
 capacityA ← capacityA + 2
 Output "park in zone A", accepted ← accepted + 1
 Else
 Output "space full", rejected ← rejected + 1
 EndIf
 EndIf
 EndIf
 B ← B + 1
 C ← C + 1
 EndIf
 V = V + 1

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EndIf
Else If category == 'S' then
    If type == 'B' OR type == 'C' then
        If capacityB < 40 then
            capacityB ← capacityB + 1
            Output "you may park in zone B", Accepted ← accepted + 1
        Else
            Output "space full", Rejected ← Rejected + 1
            If type == 'B' then
                B ← B + 1
            Else
                C ← C + 1
            EndIf
        EndIf
    EndIf
Else If type == 'V' then
    If capacityC < 14 then
        capacityC ← capacityC + 2, V ← V + 1
        Output "Park in Zone C", Accepted ← Accepted + 1
    Else
        Output "space full", Rejected ← Rejected + 1
    EndIf
EndIf
EndIf

Else if category == 'G' then
    If type == 'B' OR type == 'C' then
        If capacityC < 15 then
            CapacityC ← capacityC + 1, Accepted ← accepted + 1
            Output "Park in Zone C"
        Else
            Output "space full", Rejected ← Rejected + 1
        EndIf
    EndIf
Else if type == 'V' then
    If capacity < 14 then
        V ← V + 1
    EndIf
EndIf

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    capacityC ← capacityC + 2
    Output "Park in Zone C", Noaccepted ← accepted + 1
Else
    Output "Space full", Rejected ← rejected + 1
EndIf
EndIf
EndIf
EndIf
EndIf
Else
    Output "No valid permit"
EndIf
Output "Total Vehicles are", vehicles, "Out of which", accepted,
"were accepted and", rejected "were rejected". Out of those
accepted vehicles, b, "were bikes", c, "were cars" and, v,
"were vans."
RA = 20 - capacityA
RB = 40 - capacityB
RC = 15 - capacityC
If (RA = RB = RC = 0) then
    Output "Campus parking facility is full"
Else
    Output "Capacity for zone A: ", RA, "Zone B: ", RB,
    "Zone C: ", RC
EndIf
If CapacityA > capacityB and capacityA > capacityC then
    Output "zone A has highest occupancy"

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Else if capacity B $>$ capacity C AND capacity B $>$ capacity A then
Output "Zone B has the highest occupancy"

Else

Output "Zone C has highest occupancy"

EndIf

EndIf

→ Algorithm

Else Output "you don't have a valid permit"
EndIf

Q5.

Flowchart:





